Advanced Financial Modelling, Module 2: 3-Statement Modelling Case Study from a Blank Excel Sheet, Debt vs. Equity, and IFRS Valuation (EasyJet) (1 Week)

In this module, you'll build a combined half-year and annual 3-statement model for EasyJet starting from a blank Excel sheet, and you'll use it to make a debt vs. equity financing recommendation for the company.

Among other topics, you'll learn how to find and enter data efficiently, how to simplify and consolidate the financial statements and the key drivers, and how to build a debt and equity schedule with limited information.

You'll use the model's output to analyse the company's credit stats and ratios in different scenarios and answer the case study questions.

AFM-02-01: 1-Week Case Study from Blank Excel Sheet (EasyJet): Intro and Data Gathering (29:23)

In this introductory lesson, you'll learn how to gather data for EasyJet and set up a simple set of revenue drivers for the company's half-year periods based on their earnings results.

Table of Contents:

- **4:56:** Case Background & Data Sources
- **10:50:** Quick Conclusions and Drivers
- **16:43:** Finding and Entering the Revenue Drivers
- **25:55:** Time Skip – Formatted Version
- **27:42:** Recap and Summary

Quiz Questions:

16:42 Why is it not a good idea to use numbers such as Passengers, Seats Flown, or Sector Length in the Revenue Drivers for EasyJet?

a) It's too difficult to link all these together to create a formula for Revenue.
b) The case document strongly hints that ASK should be the key top-line driver.
c) We only have one week to complete this case study, so the Revenue and Expense Drivers must be fairly simple.
d) All of the above.

Explanation: Everything above is true. It's best to focus on simplicity in any time-limited case study, which means, at most, 3-4 main revenue drivers. For airlines, ASK or ASM always acts as the key top-line driver, and the case document makes it obvious that it's important. Going from

ASK to Passengers, Seats Flown, Sectors Flown, etc., is not worth the time and effort, especially given the extra data retrieval required.

AFM-02-02: Inputting and Simplifying the Historical Income Statement (23:25)

You'll input EasyJet's historical half-year Income Statements in this lesson, simplifying some categories and adding more detail to others.

Table of Contents:

- **2:15:** Revenue and EBITDAR
- **8:43:** Depreciation and Amortisation
- **17:31:** Financing Costs, Pre-Tax Income, and Net Income
- **20:48:** Formatting and Older Numbers (Time Skip)
- **21:57:** Recap and Summary

Quiz Questions:

17:30 In which of the following cases is it NOT worth separating Lease Depreciation from normal Depreciation on PP&E?

a) If one of them is very small next to the other.
b) If the company's Lease Liabilities are small and combined with its Debt, and its ROU Assets are also small and combined with normal PP&E.
c) If the company follows U.S. GAAP and has no or very low Finance Leases.
d) All of the above.

Explanation: Everything above is true. Lease Depreciation is worth splitting out when it's significant vs. normal Depreciation, which usually means that the company's ROU Assets are significant vs. their Net PP&E and that Lease Liabilities are a decent percentage of the company's Debt. These conditions are more likely to be true for IFRS-based companies; it is rare to split out Lease Depreciation for companies that follow U.S. GAAP because they do not record Depreciation for Operating Leases (only for Finance Leases, which are usually small).

AFM-02-03: Inputting and Simplifying the Historical Balance Sheet (23:50)

In this lesson, you'll learn how to input EasyJet's historical Balance Sheets by consolidating and simplifying some line items and listing others in more detail.

Table of Contents:

- **1:54:** Older Balance Sheets
- **11:37:** Additional Tweaks
- **14:35:** Most Recent Balance Sheet
- **20:59:** Older Numbers and Formatting (Time Skip)
- **22:39:** Recap and Summary

Quiz Questions:

14:35 What is a reasonable goal when simplifying and consolidating a company's Balance Sheet?

a) Always show the most common line items, such as Cash, AR, Inventory, PP&E, etc., separately, and combine everything else into these lines.
b) Aim for, at most, 10 items on each side, and ideally, closer to 5.
c) Ensure that each line item on the Balance Sheet links to *only one* line item on the Cash Flow Statement.
d) All of the above.

Explanation: The second answer choice is the best one. Ideally, you should aim for ~5 line items on each side of the Balance Sheet and perhaps up to ~10 for more complex companies in industries with more complicated capital structures and operations. The first answer choice is not the best because not every company has these line items; for example, software companies do not have Inventory. The third answer choice is wrong because standard line items such as PP&E often link to multiple items on the Cash Flow Statement, such as Capital Expenditures and Depreciation.

AFM-02-04: Inputting the Historical Cash Flow Statement and Making It Start with Net Income (27:13)

You'll learn how to modify EasyJet's historical Cash Flow Statements so that they start with Net Income and make adjustments based on that in this lesson; you'll also learn which line items can be consolidated and simplified.

Table of Contents:

- **1:33:** Cash Flow from Operations
- **13:46:** Cash Flow from Investing
- **15:36:** Cash Flow from Financing
- **20:34:** Checks and Other Changes
- **24:46:** Older Cash Flow Statements (Time Skip)

- **25:22:** Recap and Summary

Quiz Questions:

13:45 Which of the following is a common adjustment when "converting" a company's Cash Flow Statement to make it start with Net Income rather than Operating Income?

a) **Change "Cash Taxes" to "Deferred Taxes" so that it represents the difference between Book Taxes and Cash Taxes.**
b) Simplify the Change in Working Capital so that it's only a single line item.
c) Group together smaller line items into an "Other" category.
d) All of the above.

Explanation: The first answer is the best choice because it is required for any company with a significant difference between Book and Cash Taxes. Starting with Operating Income vs. Net Income is the difference between showing "Cash Taxes" as a lump-sum total vs. showing Deferred Taxes, representing an adjustment to Book Taxes. The other two answer choices are common adjustments and changes when inputting the historical Cash Flow Statement, but they are done regardless of what the CFS starts with, so this Operating Income vs. Net Income distinction doesn't matter.

AFM-02-05: Revenue and Expense Drivers and Income Statement Projections (29:30)

In this lesson, you'll analyse EasyJet's expense trends, project its revenue and expenses, and create an Income Statement forecast for the company based on a mix of half-year and full-year periods.

NOTE TO LESSON: In the 23:39 – 22:46 segment here, we make a mistake with the 25% tax rate. It should take effect starting in 1H 24 based on the start date in the press release (1 April 2023), so the current 19% tax rate should apply until then. We fix this problem in a later lesson.

Table of Contents:

- **3:19:** Expense Trends
- **7:00:** Revenue Drivers
- **17:36:** Expense Drivers
- **22:47:** Income Statement Projections
- **27:52:** Recap and Summary

Quiz Questions:

17:36 What's the best way to determine if our ASK projections for EasyJet are reasonable?

a) Check the company's guidance for its return to pre-pandemic capacity.
b) Review the historical growth rates in the last few "normal" years.
c) Examine the link between CapEx spending and ASK growth (later in the model).
d) All of the above.

Explanation: Everything above is true. To foot the ASK projections, you should compare them to the company's "capacity" guidance, review the historical growth rates from before the pandemic, and look at how much ASK has grown in previous years relative to the company's CapEx spending in those years. We can't check this last point at this stage because we need the CFS projections first.

AFM-02-06: Balance Sheet and Cash Flow Statement Drivers and Projections (35:50)

You'll create drivers and projections for EasyJet's Balance Sheet and Cash Flow Statement in this lesson, you'll link the financial statements, and you'll learn how to tweak the numbers afterward to get more reasonable output.

Table of Contents:

- **2:32:** Balance Sheet Drivers
- **9:55:** Cash Flow Statement Drivers
- **27:02:** Linking and Tweaking the Numbers
- **34:00:** Recap and Summary

Quiz Questions:

34:00 Even if our CapEx estimates in the projected years match the company's estimates, why might our financial model still have problems?

a) The cash flow might still be off if we're assuming higher or lower revenue growth based on that CapEx.
b) The Depreciation figures could be off, which means the cash flows will be distorted due to the CapEx – Depreciation net figure.
c) In a non-annual model, the timing of the CapEx spending might not match the company's (e.g., differences between Q1-Q2 and Q3-Q4).
d) All of the above.

Explanation: Everything above is true. Even if the CapEx numbers match the company's estimates in absolute currency terms, the cash flows might still be off because the revenue

assumptions or the Depreciation figures may not match up. Specifically, the company might be assuming more or less revenue growth from the CapEx spending than we are, and the gap between Depreciation and CapEx might also be different. The timing issue is also possible, but it's less of a problem in most models; timing differences within a specific year matter more if lenders evaluate the company's credit stats and ratios frequently.

AFM-02-07: Debt & Equity Schedule, Part 1: Required Funding (21:55)

In this lesson, you'll calculate EasyJet's required funding or excess liquidity to determine how much Debt or Equity it must issue or how much Debt it can repay in each period; you'll also learn how to find and enter information on the existing Debt tranches, maturities, and amortisation from the filings and case document.

NOTE TO LESSON: In the 15:03 – 15:08 segment here, we forget to include Dividends in the calculation. We fix this problem later in the lesson, in the 21:02 – 21:10 segment.

Table of Contents:

- **1:39:** Part 1: Current Debt Tranches
- **9:18:** Part 2: Mandatory Repayments and Maturities
- **15:41:** Part 3: Minimum Cash and Funding Required
- **20:24:** Recap and Summary

Quiz Questions:

20:24 What is the biggest drawback of linking EasyJet's Minimum Cash balance to its Deferred Revenue?

a) Deferred Revenue won't necessarily trend with the company's capacity (ASK) or Revenue.
b) Historically, Deferred Revenue as a % of Revenue has spanned a huge range of values.
c) Deferred Revenue represents what the company has collected upfront in cash from customers without delivery – not what it owes.
d) Most likely, the company will put in place another Revolver for additional liquidity beyond its Cash balance.

Explanation: The second answer choice is the biggest issue here. Since Deferred Revenue as a percentage of Revenue has fluctuated so much historically, it's not necessarily clear what it should be going forward. Deferred Revenue trends with Revenue and ASK, so the first answer is incorrect. The third answer is a correct statement, but it doesn't answer the question because this is not a real drawback of defining the Minimum Cash this way (as Deferred Revenue also

represents a certain amount of future costs required for product/service delivery). The last answer is not the best choice because the presence of a Revolver changes the Minimum Cash formula a bit, but it doesn't alter the main components.

AFM-02-08: Debt & Equity Schedule, Part 2: Issuances and Repayments (20:04)

You'll learn how to estimate EasyJet's additional Debt and Equity issuances and Debt Repayments in this lesson, including the "order of operations" for its different capital sources.

NOTE TO LESSON: In the 7:50 – 8:05 segment, we suggest drawing on the Eurobonds first because their current weighted average interest rate is lower than the rate on the Term Loans. We change this assumption later due to rising rates, and by the end of the entire model, we also change the "draw order."

Table of Contents:

- **1:42:** Part 1: Term Loan Repayments
- **4:47:** Part 2: Debt vs. Equity Split
- **7:36:** Part 3: Debt and Equity Draws
- **12:47:** Part 4: Links and Tests
- **17:48:** Recap and Summary

Quiz Questions:

12:47 When determining the available capacity on EasyJet's Term Loans and Eurobonds, why do we add the maturities and principal amortisation?

a) Because these represent cash outflows, which always increase the capacity of these facilities.
b) Because the maximum capacity for each tranche of Debt keeps increasing with the maturities and amortisation.
c) Because these items reduce the amount of drawn Debt, which means the company can now draw on more Debt in the future.
d) All of the above.

Explanation: The third answer choice is the correct one. Any type of Debt principal repayment, whether in the form of scheduled amortisation or fixed maturities, reduces the amount of drawn Debt in a specific facility, which means the company can now draw on more Debt. The first answer makes no sense because cash outflows alone do not increase the capacity of these facilities, and the second answer is wrong because the maximum capacity for each tranche stays the same; all that changes is how much is currently drawn on the facility.

AFM-02-09: Debt & Equity Schedule, Part 3: Net Interest Expense and Links (22:25)

In this lesson, you'll finish the Debt & Equity Schedule by researching the interest rates on the different Debt tranches, calculating the Net Interest Expense, and linking these calculations to the rest of the model.

Table of Contents:

- **2:23:** Part 1: Estimating the Interest Rates
- **14:07:** Part 2: Calculating the Net Interest Expense
- **17:13:** Part 3: Linking the Schedule to the Statements
- **20:29:** Recap and Summary

Quiz Questions:

14:06 Which of the following interest rates is LEAST likely to change over time as central banks increase or reduce rates?

a) The Discount Rate used to calculate Lease Interest.
b) The floating interest rate on a Term Loan facility.
c) The coupon rate on a Eurobond facility that allows for issuances at different times and terms.
d) The coupon rate on a Eurobond facility that is already drawn at maximum capacity.

Explanation: The last answer choice is the best one. "Bonds" tend to have fixed coupon rates that do not change over time, even as overall interest rates do, so if a Eurobond facility is already drawn to its maximum level, its interest rate is unlikely to change anytime soon (unless there is a maturity). The floating interest rate on a Term Loan will always change over time, and the Discount Rate used in the Lease calculations will also change, but usually to a lesser extent than the rates on actual Debt.

AFM-02-10: Key Metrics and Ratios (ROCE, Leverage Ratio, and Coverage Ratio) (21:03)

You'll calculate the key metrics and ratios for EasyJet in this lesson, including the Return on Capital Employed (ROCE), the Leverage and Coverage Ratios, the Debt / Total Capital Ratio, and variations on these, and you'll draw some quick conclusions about the company's future financing.

Table of Contents:

- **2:38:** Which Metrics to Include
- **8:08:** ROCE Calculation
- **12:13:** Other Calculations
- **16:27:** Interpretation
- **19:18:** Recap and Summary

Quiz Questions:

12:13 What is the main PROBLEM with defining Return on Capital Employed as EasyJet has done (Operating Income / Average Capital Employed)?

a) We should use the LTM Operating Income, not the number in just one half-year period.
b) Average Capital Employed should be based on annual averages rather than the average over a half-year period.
c) Operating Income deducts part of the Lease Expense, but the Lease Liabilities in Capital Employed correspond to the entire future expense.
d) All of the above.

Explanation: The third answer choice is the correct answer. If the denominator of a metric such as ROCE adds an entire Liability, the numerator should exclude or add back the full expense associated with that Liability. But Operating Income under IFRS deducts only the Lease Depreciation, not the Lease Interest, so part of the Lease Expense here is excluded or added back, and the other part is deducted, which is inconsistent. The first two answers are wrong because it's not necessary to use the LTM number since we're looking at the ROCE in one specific period, and annual averages make sense only if we're also using annual rather than half-year NOPAT figures.

AFM-02-11: Annual Summaries (27:43)

In this lesson, you'll add annual summaries to the EasyJet model and use them to find and fix several problems with the Dividend calculations and the Tax Rate; you'll also learn how these summaries can highlight assumptions that might need to be tweaked.

Table of Contents:

- **2:17:** Setup for Revenue Drivers
- **7:57:** Expense, BS, and CFS Drivers
- **9:14:** Income Statement
- **10:44:** Cash Flow Statement
- **13:19:** Balance Sheet

- **16:20:** Drivers and Fixes
- **22:34:** Key Metrics and Ratios
- **25:39:** Recap and Summary

Quiz Questions:

22:34 Which of the following is an ADVANTAGE of creating annual summaries in a quarterly or half-year model?

a) It's easier to see high-level trends at a glance.
b) It's easier to compare the model to the company's internal guidance.
c) It's easier to find errors and oversights in the non-annual assumptions.
d) All of the above.

Explanation: Everything above is true. These all represent the advantages or benefits of the annual summaries in a non-annual model, as it's often difficult to interpret quarterly and half-year models without some type of summary.

AFM-02-12: Operating Scenarios and Growth and CapEx Adjustments (25:39)

You'll tweak the baseline revenue, capacity growth, and CapEx assumptions to get more reasonable numbers in this lesson, and you'll extend this approach to build Downside and Extreme Downside cases to assess the company's credit stats and ratios with a much slower recovery to normal.

Table of Contents:

- **1:19:** Baseline Revenue and CapEx
- **7:41:** Scenario Setup
- **14:49:** Downside and Extreme Downside Scenarios
- **23:50:** Recap and Summary

Quiz Questions:

23:49 How should the Downside and Extreme Downside cases in a model differ from the Base case?

a) Revenue growth and "capacity" should be lower.
b) Operating margins should be lower.
c) The company should require more outside financing to continue operating.

d) **All of the above.**

Explanation: Everything above is true. In pessimistic scenarios, the growth rates and margins should both be lower, which means the company's cash flow will be worse, and it will be more dependent on external sources of capital.

AFM-02-13: Net Operating Losses (NOLs), Deferred Taxes, and Model Tweaks (24:40)

In this final Excel-based lesson, you'll build Net Operating Losses into the model, tweak the Debt & Equity Schedule to use more reasonable draw and interest rate assumptions, and add some labels and titles/headers.

Table of Contents:

- **2:32:** NOLs and Deferred Taxes
- **9:55:** Debt & Equity Schedule Adjustments
- **20:23:** Other Model Tweaks
- **22:57:** Recap and Summary

Quiz Questions:

9:54 When is it useful to include Net Operating Losses in a 3-statement financial model?

a) When the company has a high NOL balance due to losses in previous years.
b) When the company provides detailed information on the components of its Deferred Tax Asset (DTA).
c) **When the company's NOL balance is such that it will significantly affect its Cash Taxes over the entire projected period.**
d) All of the above.

Explanation: The third answer choice is correct. NOLs are worth including if they're expected to impact a company's Cash Taxes significantly for many years into the future. Even if the NOL balance is "high," as in the first answer choice, it doesn't mean much if the company also uses up the entire balance quickly, resulting in minimal Cash-Tax impact over the projected period. Detailed information about the DTA components, as in answer choice #2, also makes little difference if the NOLs expire or get used up quickly without significantly impacting the Cash Taxes.

AFM-02-14: Case Study Answers, Grading Criteria, and the Time Limit (29:05)

You'll use the final Excel model to answer the case study questions in this lesson, and you'll learn about what makes answers to this type of open-ended case study "correct" or "not correct" and how to interpret your performance and completion time vs. the 1-week time allowance.

Table of Contents:

- **0:55:** ROCE and EBITDAR Margins
- **11:03:** Debt vs. Equity
- **20:01:** Other Compliance Methods
- **24:56:** Grading Criteria
- **27:24:** Time Limit and Performance

Quiz Questions:

20:01 Suppose that a company can comply with its maximum Leverage Ratio and minimum Interest Coverage Ratio with 60% Debt and 40% Equity funding. Should you recommend these exact numbers?

a) **It depends on how much of a "cushion" there is with these numbers.**
b) Yes, because the company can adjust its financing as needed if its business results change.
c) No, because there needs to be at least a 20-30% "cushion" with these targets – simple compliance is not enough.

Explanation: The first answer choice is the best one. Generally, you want at least a bit of a "cushion" when recommending a Debt and Equity funding mix. For example, if the company's Debt / EBITDA is 5.9x against a maximum of 6.0x, that's not much of a cushion, and you should probably recommend a slightly lower Debt percentage. Numbers in the 5.0 – 5.5x range would be more acceptable. The second answer is not the best choice because, although this is true to some extent, the company cannot "undo" financing decisions it made in previous periods; once the capital is raised, it is raised. This answer choice is more applicable in cases where the company's performance varies primarily in 1-2 periods near the end of the projections. The third answer is not the best choice because it doesn't have to be a 20-30% cushion, specifically; just something more than "right at or below the maximum level" for a given credit stat.

AFM-02-15: BONUS: IFRS-Based Valuation and DCF for EasyJet and Debt vs. Equity Presentation [OPTIONAL] (29:06)

In this bonus, optional lesson, you'll get a full valuation and DCF for EasyJet and a debt vs. equity recommendation in slide format. These additional materials give you more templates and examples to draw on, especially if you are valuing IFRS-based companies instead of ones that follow U.S. GAAP.

NOTE TO LESSON: The audio at 11:52 should be "the CapEx per ASK here," not "the ASK here."

Table of Contents:

- **2:17:** Part 1: Public Comps
- **6:41:** Part 2: Precedent Transactions
- **8:18:** Part 3: Discount Rate (WACC)
- **10:03:** Part 4: DCF Setup and Issues with Leases and NOLs
- **17:24:** Part 5: Valuation Summary
- **19:38:** Part 6: Debt vs. Equity and Valuation Presentation
- **27:42:** Recap and Summary

Quiz Questions:

17:23 What is the BIGGEST difference in a DCF when you are valuing a company that follows IFRS rather than U.S. GAAP?

a) You must account for NOLs differently by building them into the cash flow projections.
b) It's more important to use a different TEV bridge in the DCF than in the Public Comps.
c) Leases are much more significant, and you must either deduct or add back/exclude the entire Lease Expense.
d) All of the above.

Explanation: The third answer is the best choice. NOLs do not work differently under IFRS: you still either factor them into the cash flow projections or include them in the TEV bridge, so the first answer is wrong. The second answer is wrong because it's not "more important" to use a different TEV bridge in different parts of the valuation; it's just that you may have to do this depending on how you treat Lease Liabilities in the analysis. The third answer is the best choice: since both the Operating Lease and Finance Lease Expenses are split into Interest and Depreciation under IFRS, the impact of Leases is far more significant, and you have to be careful to deduct or exclude/add back the entire expense.